

README.md

SynChain

AI-Powered Supply Chain Decision Intelligence Platform

Overview

SynChain is a Multi-Agent AI Supply Chain Digital Twin platform designed to help businesses simulate supply chain decisions before applying them in the real world.

The platform creates a virtual representation of supply chain operations and uses specialized AI agents to analyze demand, inventory, logistics, and risk factors.

Current Status

Implemented

- Next.js Frontend
- FastAPI Backend
- SQLite Database
- Simulation Engine
- Results Dashboard
- Multi-Agent Architecture Foundation

In Development

- Digital Twin State Engine
- Agent Orchestrator
- Scenario Simulation Engine

Future Vision

- Real-Time External Intelligence
- Industry News Monitoring
- Predictive Procurement
- Enterprise Analytics

Core Value

SynChain helps organizations answer:

- How much inventory should we keep?
- Which warehouse should fulfill demand?

- What risks exist in the supply chain?
- What happens if conditions change?

Architecture Summary

User → Frontend → API → Agent System → Decision Engine → Dashboard

Technology Stack

Frontend:

- Next.js
- React
- TailwindCSS

Backend:

- FastAPI
- SQLAlchemy

Database:

- SQLite (current)
- PostgreSQL (planned)